



UKUDLA Data Science Certificate

Descriptive Analysis

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Visual patterns need numerical descriptions

Visualization helps us see

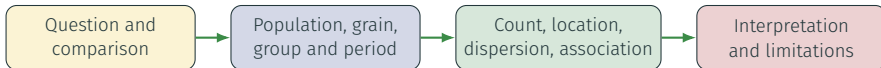
- distributions and unusual values;
- differences between treatments, sites or groups;
- changes over time; and
- possible associations.

Description helps us state

- how many observations contribute;
- what is typical and how values vary;
- how groups and periods differ; and
- how strongly variables vary together.

Descriptive statistics compress observed data into inspectable quantities; they complement rather than replace plots and individual observations.

A summary depends on population, group and time



- The same mean can conceal different variation or skewness.
- A pooled summary can conceal group, batch, site or period differences.
- Missing values change the effective number of observations.
- A relationship observed in one period may not persist in another.

This session moves from **motivation**, through essential **concepts**, to a reproducible **application** for analytical data.

Begin with the observations being described

State the scope

- target population and observed sample;
- observation grain and candidate key;
- categorical and numerical variables;
- groups, periods and weighting; and
- missing-value rule.

Report the denominator

- total rows;
- non-missing observations;
- counts by relevant group and period;
- frequencies and proportions; and
- observations excluded from a statistic.

A statistic without its population, grouping and effective sample size is incomplete.

Describe location, dispersion and shape together

Feature	Measures	Interpretation and caution
Location	Mean, median, quantiles	Mean uses every value; median is less sensitive to extremes
Dispersion	Range, IQR, variance, standard deviation	Range depends on extremes; SD shares the variable's unit
Shape	Skewness, tails, modes, unusual values	Needs visual inspection; an unusual value is not automatically an error
Coverage	Count, missing count, proportion available	The effective sample can differ between variables and groups

Use a small set of complementary summaries. A single average cannot represent distribution, variability, coverage and shape.

Association measures quantify co-variation

Covariance

- sign indicates joint direction;
- magnitude depends on both units;
- sensitive to extreme observations; and
- difficult to compare across scales.

Pearson correlation

- standardizes linear co-variation;
- ranges from -1 to 1 ;
- can hide nonlinear patterns; and
- changes with groups, periods and outliers.

Always pair association measures with a plot and the contributing observation count.
Correlation is neither explanation nor causation.

Stationarity asks whether the process remains stable

Descriptive intuition

Across time, ask whether:

- the distribution shifts;
- the typical level changes;
- variability changes; and
- associations change.

Weak stationarity

A time process has:

- a constant mean;
- a constant variance; and
- covariance depending on lag, not calendar time.

Stationarity is an assumption about a data-generating process. A finite dataset can provide evidence for or against it, but cannot prove it universally.

Look for evidence of change before modeling

Pattern	Descriptive evidence	Why later models care
Trend	Period means or medians differ systematically	A constant-mean model is inappropriate
Changing variance	Period SDs or IQRs differ	Errors and uncertainty may not be stable
Structural change	An abrupt level or relationship shift	Historical parameters may not transfer
Seasonality/cycles	Repeated time-linked patterns	Time dependence remains in observations
Association drift	Correlations differ by group, site or period	One pooled slope may conceal changing relationships

Compare plots and grouped summaries across meaningful periods. Formal stationarity tests are optional extensions, not automatic verdicts.

Describe an outcome by group and period

For every group, report

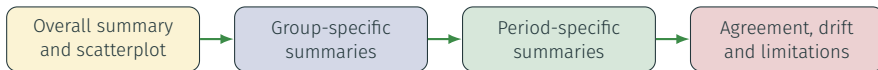
- total and non-missing n ;
- mean and median outcome;
- quartiles and IQR;
- standard deviation and range; and
- first and last occasion represented.

Compare periods

- an earlier historical period;
- a recent development period;
- a later evaluation period;
- levels, dispersion and coverage; and
- agreement with trend graphics.

Period summaries make potential distribution shift visible before a model is fit.

Describe relationships without pooling blindly



- Report each variable's location, dispersion, range and missingness.
- Calculate covariance and correlation with their contributing n .
- Compare pooled, group-specific and period-specific associations.
- Investigate whether between-group differences drive the pooled result.
- Retain measurement, aggregation and non-causal interpretation boundaries.

Key message: describe stability before modeling

Reproducible artifacts

- descriptive-analysis script;
- group and period summary tables;
- association table with effective n ;
- compact report with linked figures; and
- explicit grouping and missingness rules.

Modeling hand-off

- evidence of trends or level shifts;
- evidence of changing variability;
- stability of associations;
- implications for time-based splitting; and
- questions requiring a stated model.

Description does not fix non-stationarity. It reveals when later explanatory and predictive models must represent time, change or limited transferability.

Summarize what was observed—and expose what may not remain stable.

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